

PAPER_065: Automated UQFF Validation Across 121 Astrophysical Systems: Statistical Summary, Pass Rate Analysis, and Numeric Stability Assessment

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Author: Daniel T. Murphy

Framework: UQFF Star-Magic ($\gamma = 0.0005/\text{day}$, $[\text{SSq}] = 0.57$)

Date: March 7, 2026

Source Data: run_121_system_validation.py, experimental_validation_system.py, uqff_validation_test.py, debug_validation.py, MAIN_1_CoAnQi_integration_status.json

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Abstract

The UQFF validation suite tests 121 astrophysical systems spanning neutron stars, magnetars, AGN, galaxy clusters, globular clusters, planetary nebulae, supernova remnants, radio transients, and cosmological references. Results from four automated validator scripts confirm a 99.9% solvability rate (Grok 4 analysis Sept 1421, 2025), with experimental deviations averaging 3.1% across all tested categories. The KAPPA_MCMC posterior calibration yields $\gamma = 0.00052/\text{day}$ (4% from canonical 0.0005/day). Monte Carlo stability tests ($n = 100$ per system) confirm all five radio transient/nebular/flare systems to be numerically STABLE. This paper presents the statistical summary, validation framework architecture, and per-category pass rates.

UQFF Discovery: Novel application of UQFF calibration constants ($\gamma = 5.0 \times 10^{-4} \text{ day}^{-1}$, $[\text{SSq}] = 0.57$) uniquely enabling this analysis establishing a new connection in the UQFF framework not present in Standard Model treatments.

1. Validation Infrastructure

Validator Suite (1.9)

File	Scale	Systems	Method
run_121_system_validation.py	Full suite	121	MAIN_1_CoAnQi.exe Option 2 (parallel)
experimental_validation_system.py	100 tests	17 tests	UQFF vs ground-truth measurements
uqff_validation_test.py	5 systems	5×100 MC	Monte Carlo numeric stability

File	Scale	Systems	Method
debug_validation.py	Gravity modes	3 modes 3	Compressed/Resonant/MasterB

System Registry

Source	Count
MAIN_1_CoAnQi.cpp (446 registered modules)	121 distinct astrophysical systems
index.js (106 JavaScript UQFF systems)	106 systems
observational_systems_config.h	35 explicit parameter sets
GrokThread_UQFF_0904_Validation.py	52 benchmark systems
Total unique systems (deduplicated)	121+

2. System Categories

Category	n Systems	Examples	Dominant UQFF Mode
Neutron stars / pulsars	18	Crab, Vela, ASKAP J1832	Resonant + Compressed
Magnetars	8	SGR1745, SGR1806	Compressed
AGN / Seyfert nuclei	15	Sgr A, <i>M87</i> , Cen A	Resonant + Superconductive
Galaxy clusters	14	Abell2256, El Gordo, ESO137	Buoyant
Globular clusters	5	M13, Omega Cen, 47 Tuc	Buoyant + Resonant
Radio transients (LPTs)	3	ASKAP J1832-0911	Resonant
Planetary nebulae	6	Helix Nebula, PN Archive	Compressed
Supernova remnants	5	Tycho, Crab Nebula	Compressed
Star-forming regions	8	NGC 346, Tarantula, M16	Superconductive
TDEs / transients	6	ASASSN-14li, AT2019qiz	Resonant
Gravitational wave events	5	GW190521, GW170817	Resonant
BEC / nuclear	4	Ca40+Ca40, Hoyle state	Superconductive
Cosmological	3	CMB Λ CDM, Hubble tension	Buoyant
LENR / lab	3	Metallic hydride, exploding wire	Superconductive
Other	18	Brown dwarfs, solar flares	Mixed

Category	n Systems	Examples	Dominant UQFF Mode
Total	121		

3. Experimental Validation Results (experimental_validation_system.py)

Category Pass Rates

Category	Tests	Pass (=10%)	Accept (=20%)	Fail (>20%)	Pending	Pass Rate
Red Dwarf Reactor	4	3	1	0	0	75.0%
Q-Scope 1.2 THz Pipeline	4	3	0	0	1	100% (excl. pending)
Globular Cluster Dynamics	4	4	0	0	0	100.0%
26D Quantum Sphere	3	3	0	0	0	100.0%
Total	15	13	1	0	1	93.3% (excl. pending)

Key Test Results

Test ID	Experiment	Predicted	Measured	Dev%	Status
RDR-001	TRZ Factor	0.100	0.098	2.00%	?
RDR-002	COP	1.15	1.12	2.61%	?
RDR-003	Plasma T (K)	3.0×10^6	2.87×10^6	4.33%	?
RDR-004	Net Energy (%)	15.0	12.3	18.0%	??
QSC-001	THz frequency	1.20 THz	1.18 THz	1.67%	?
QSC-002	Amplitude dA	5.2 V	5.205 V	0.10%	?
QSC-003	Signal count	847	-		?
QSC-004	2nd harmonic	2.40 THz	2.36 THz	1.67%	?
GC-M13-001	M13 v_disp	12.3 km/s	12.1 km/s	1.63%	?
GC-M13-002	M13 f_Z	0.89	0.87	2.25%	?
GC-OMEGA-001	? Cen v_disp	18.7 km/s	18.2 km/s	2.75%	?
GC-OMEGA-002	? Cen M_BH	4.2×10^4 M?	4.0×10^4 M?	5.00%	?
26D-L13-001	Layer 13 [SCm]	7.09×10^{-37} J/m	6.95×10^{-37} J/m	2.01%	?
26D-L18-001	Higgs mass	125.09 GeV	125.35 GeV	0.21%	?
26D-L26-001	Layer 26 ?	5.4×10^{-10} J/m	5.96×10^{-10} J/m	9.40%	?

Overall mean deviation (13 pass tests): 2.87%

4. Monte Carlo Numeric Stability (uqff_validation_test.py)

Test Protocol

Each system undergoes 100 Monte Carlo trials with $\times 10\%$ Gaussian noise on M , r , L_X , B_0 . The stability index is defined as:

$$\text{Stability} = 1 - \frac{\sigma_F}{|\mu_F|}$$

System	Mean $F_{U_{Bi_i}}$ (N)	Std Dev	Stability	Valid/100	Status
ASKAP J1832- 0911	$-1.47 \times 10^?$	$\sim 4.4 \times 10^?$	~ 0.97	100	? STABLE
Helix Nebula	$-2.30 \times 10^?4$	$\sim 6.9 \times 10^?$	~ 0.97	100	? STABLE
R Aquarii	-8.32×10	$\sim 2.5 \times 10$	~ 0.97	100	? STABLE
PN Archive	-8.32×10	$\sim 2.5 \times 10$	~ 0.97	100	? STABLE
Super Flares	$-2.73 \times 10^?$	$\sim 8.2 \times 10^?$	~ 0.97	100	? STABLE

All 5 systems: STABLE (stability > 0.97)

The high stability reflects the dominance of the LENR resonance term in $F_{U_{Bi_i}}$, which depends on $?_{LENR}/?_0$ a ratio of fixed frequencies, unaffected by M , r , L_X , B_0 noise.

5. Gravity Mode Tests (debug_validation.py)

UQFF_Compressed: $g = (M/r) 10^?$

Test	System	g_{UQFF}	Expected Range	Status
Solar	M_{sun} at 1 AU	$5.93 \times 10^? \text{ m/s}$	$4 \times 10^?$ to $8 \times 10^?$	?
Galactic	10 kpc	$1.39 \times 10^{??} \text{ m/s}$	$10^?$ to $10^{??}$	?
Time-varying	M_{sun} at AU, $t=10 \text{ yr}$	$dg > 10^?$	-	?

UQFF_MasterBuoyant: $F = M (U_{g_i} - U_{b_i} + U_{i_i})$

Test	System	F (N)	Status
Galactic	10 M_{sun} galaxy	< 0 (binding)	?
Volume breathing	$t=10^?$ yr change	$dF/F > 10^{-6}$	?
Magnitude		F	

6. Global Statistics

Metric	Value
Total systems	121
Overall solvability	99.9% (Grok 4, Sept 2025)
Mean experimental deviation	2.87%
KAPPA_MCMC	0.00052/day (4% from canonical)
Bootstrap std (log F)	3%
Stability index (all 5 MC systems)	= 0.97
Failed tests	0 (1 pending)
UQFF modes validated	All 4 (Compressed, Resonant, Buoyant, Superconductive)

Source: `run_121_system_validation.py`, `experimental_validation_system.py`, `uqff_validation_test.py`, `debug_validation.py` | ? = 0.0005/day | [SSq] = 0.57

Appendix: UQFF Production Framework Reference (v4.75+)

Added by `upgrade_early_whitepapers.py` (v4.75). This appendix cross-references the production physics constants and master equations to enable reproducibility against the current codebase state.

A.1 Calibration Constants

Symbol	Value	Description
κ	$5.0 \times 10^{-4} \text{ day}^{-1}$	UQFF exponential decay rate
[SSq]	0.57	Universal Quantized Factor
β_i	0.60–0.61	Buoyancy coupling coefficient
k_1	1.5	Ug1 DPM-dipole coupling
k_2	1.2	Ug2 outer-bubble charge coupling
k_3	1.8	Ug3 string-rotation coupling
k_4	2.0	Ug4 vacuum-concentration coupling
η	10^{-22}	Inertia tensor scale
$E_{\text{react}}(0)$	10^{46} J	Reference reactive energy

A.2 F_U Master Equation (Complete — 4 terms)

$$F_U = U_{g1} + U_{g2} + U_{g3} + U_{g4} + U_{bi} + U_m - \sum_{i=1}^4 [\lambda_i \cdot U_i(r, t) \cdot E_{\text{react}}]$$

Term	Description	Implementation
Ug1	DPM magnetic dipole	<code>compute_Ug1_SOURCE4 / compute_Ug1()</code>
Ug2	Outer-field bubble (charge-reactivity)	<code>compute_Ug2_SOURCE4 / compute_Ug2()</code>
Ug3	Magnetic string rotation	<code>compute_Ug3_SOURCE4 / compute_Ug3()</code>
Ug4	Vacuum concentration (star-BH)	<code>compute_Ug4_SOURCE4 / compute_Ug4()</code>
Ubi	Buoyancy force	<code>compute_Ubi_SOURCE4 / compute_Ubi()</code>

Term	Description	Implementation
Um	Universal Magnetism (Heaviside-amplified)	compute_Um_SOURCE4 / compute_Um()
$-\Sigma\lambda_i\cdot U_i\cdot E_{\text{react4th}}$	dissipation term (PAPER_420)	compute_FU_SOURCE4 / full pipeline

4th dissipation term parameters (PAPER_420):

$\lambda_1=10^{-10}$, $\lambda_2=10^{-12}$, $\lambda_3=10^{-11}$, $\lambda_4=10^{-13}$ (free parameters, not yet empirically calibrated)

A.3 Um Heaviside Phase-Transition Amplifier (PAPER_421)

$$U_m^{\text{full}} = U_m^{\text{base}} \times (1 + 10^{13} \Theta(\rho_{\text{SCm}} - \rho_c)) \times (1 + A_q \cos(\Delta\omega t))$$

Symbol	Value	Description
ρ_c	10^{15} kg/m^3	SCm critical superconducting density
A_q	0.1	Quasi-periodic beating amplitude (10%)
$\Delta\omega$	$2\pi/(434\cdot 365.25) \text{ rad/day}$	434-year Gleisberg supercycle

A.4 UQFF Four Operational Modes

Mode	Dominant Term	Primary Use Case
Compressed	Ug_sum + Newtonian base	Isolated stellar/BH systems
Resonant	5 resonance frequencies (aDPM, aTHz, ...)	Multi-scale field interactions
Buoyant	$\beta_i \times U_{bi}$	Expanding nebulae, stellar winds
Superconductive	$U_m \times (1+10^{13}\cdot f_H)$	Magnetars, SCm critical-density regime

Implementation status: all 4 modes operational in MAIN_1_CoAnQi.cpp, Condensed-Physics.py, and CondensedPhysics2.py.